

Partial Differential Equations

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Part I

Sobolev spaces

Chapter 1

Distribution theory

1.1 Space of test functions

1.1. (a) If a test function φ satisfies $\langle 1, \varphi \rangle = 0$, then there is $v \in \mathbb{R}^d$ and a test function ψ such that $\varphi = v \cdot \nabla \psi$.

(b) If a distribution has zero derivative, then it is a constant.

1.2 (Weak* convergence).

1.2 Space of distributions

1.3 (Rigged Hilbert space).

1.3 Well-posedness

1.4 (Extension of linear operators). Let $T : \mathcal{D} \rightarrow \mathcal{D}'$ be a continuous linear operator. We can always define the adjoint $T^* : \mathcal{D} \subset \mathcal{D}'' \rightarrow \mathcal{D}'$. The most reasonable extension of T is $T : (T^*(\mathcal{D}))' \rightarrow \mathcal{D}'$. For $f \in (T^*(\mathcal{D}))'$, we can define $\langle T(f), \varphi \rangle := \langle f, T^* \varphi \rangle$ for $\varphi \in \mathcal{D}$.

Suppose $T : (\mathcal{D}, \mathcal{T}) \rightarrow (T(\mathcal{D}), S)$ is proved to be continuous. If $(\mathcal{D}, \mathcal{T}) \rightarrow (T^*(\mathcal{D}))'$ and $(T(\mathcal{D}), S) \rightarrow \mathcal{D}'$ are embeddings, then the extension of T to the completion of $(\mathcal{D}, \mathcal{T})$ agrees with $T : (T^*(\mathcal{D}))' \rightarrow \mathcal{D}'$.

For example, if Φ is locally integrable, then since $(T_\Phi)^* = T_{\tilde{\Phi}}$ and $\Phi * \varphi \in \mathcal{E} = C^\infty$ for $\varphi \in \mathcal{D}$, the convolution operator $T_\Phi : \mathcal{E}' \rightarrow \mathcal{D}'$ can be defined on the space of compactly supported distributions.

If $g * f$ is well-defined, is $f * g$ also well-defined? In other words, if $f \in (T_{\tilde{g}}(\mathcal{D}))'$ so that $g * f \in \mathcal{D}'$, then $g \in (T_{\tilde{f}}(\mathcal{D}))'$? Are they same?

$$\langle g, \tilde{f} * \varphi \rangle =$$

Exercises

Chapter 2

Sobolev inequalities

2.1 Approximations

2.1 (Completeness of Sobolev norms).

2.2 (Difference quotient).

2.3 (Interior approximation).

2.4 (Myers-Serrin theorem).

2.2 Extensions and restrictions

2.5 (Lipschitz boundary).

2.6 (Extension theorem).

2.7 (Trace theorem).

2.8 (Vanishing at boundary). zero trace, whole domain

2.3 Sobolev embeddings

Temporarily we define a *function space* on $\mathbb{R}^d$ as a complete topological vector space X together with embeddings $S(\mathbb{R}^d) \rightarrow X$ and $X \rightarrow S'(\mathbb{R}^d)$. If $S(\mathbb{R}^d)$ is dense in X , hence so is X in $S'(\mathbb{R}^d)$, we will say X is *approximable*. We will not take dual spaces for non-approximable spaces, such as $L^\infty(\mathbb{R}^d)$ and $M(\mathbb{R}^d)$.

Let X, Y be function spaces on $\mathbb{R}^d$ such that X is approximable. We claim that if $\|u\|_Y \lesssim \|u\|_X$, then we have embedding $X \subset Y$. Let $u \in X$. Since S is dense in X , we can take a net $u_\alpha \in S$ such that $u_\alpha \rightarrow u$ in X . Then, u_α is Cauchy in Y by the inequality, we have $v \in Y$ such that $u_\alpha \rightarrow v$ in Y . The uniqueness of limits in S' implies that $u = v$, hence $u \in Y$.

$$S' = \bigcup_{\alpha, \beta \in \mathbb{Z}_{\geq 0}^d} \langle x \rangle^{-\alpha} \langle \xi \rangle^{-\beta} L^2.$$

2.9. We introduce the *Sobolev regularity* $s/d - 1/p$ for a triple of $(s, d, p) \in \mathbb{R} \times \mathbb{Z}_{>0} \times [1, \infty]$, and the *Hölder regularity* $(k + \alpha)/d$ for a triple $(k, \alpha, d) \in \mathbb{Z}_{\geq 0} \times [0, 1) \times \mathbb{Z}_{>0}$.

2.10 (Gagliardo-Nirenberg-Sobolev inequality). for $(p, d) \in [1, \infty] \times \mathbb{Z}_{>0}$ for $q \in [1, \infty]$,

(a) If $p < d$ and $-1/q = 1/d - 1/p$, then

$$\|u\|_{L^q(\mathbb{R}^d)} \lesssim \|u\|_{W^{1,1}(\mathbb{R}^d)}, \quad u \in C_c^\infty(\mathbb{R}^d).$$

Proof. (a) We first show $W^{1,1} \rightarrow L^q$ for $1 < d$, where $q := \frac{d}{1-d}$, using the averaging argument by Gagliardo. Fix $u \in C_c^\infty(\mathbb{R}^d)$. Define a smooth bounded functions F_κ on $\mathbb{R}^d$ for $\kappa \subset \{1, \dots, d\}$ such that

$$F_\kappa(x) := \left(\int_{\mathbb{R}^{|\kappa|}} |\nabla u(x)| dx_\kappa \right)^{\frac{1}{d-1}}, \quad x \in \mathbb{R}^d.$$

Then,

$$|u(x)| \leq \int_{\mathbb{R}} |\partial_i u(x)| dx_i \leq \int_{\mathbb{R}} |\nabla u(x)| dx_i = F_i(x)^{d-1}, \quad x \in \mathbb{R}^d, \quad 0 \leq i \leq d,$$

implies $|u|^q \leq F_1 \cdots F_d$. Since F_κ is constant along the direction x_i if $i \in \kappa$, the Hölder inequalities get

$$\int_{\mathbb{R}} |u|^q dx_1 \leq F_1 \int_{\mathbb{R}} F_2 \cdots F_d dx_1 \leq F_1 \prod_{i=2}^d \left(\int_{\mathbb{R}} F_i^{d-1} dx_1 \right)^{\frac{1}{d-1}} = F_1 F_{12} \cdots F_{1d},$$

and

$$\int_{\mathbb{R}^2} |u|^q dx_{12} \leq F_{12} \int_{\mathbb{R}} F_1 F_{13} \cdots F_{1d} dx_2 \leq F_{12}^2 F_{123} \cdots F_{12d},$$

which continues inductively and leads finally

$$\int_{\mathbb{R}^d} |u|^q dx \leq F_{1 \dots d}^d = \left(\int_{\mathbb{R}^d} |\nabla u(x)| dx \right)^q,$$

so

$$\|u\|_{L^q(\mathbb{R}^d)} \lesssim \|u\|_{W^{1,1}(\mathbb{R}^d)}.$$

Now we prove $W^{1,p} \rightarrow L^q$ for $1 < p < d$ and $q := \frac{d-p}{p}$. If we choose $\gamma > 0$ such that

□

2.11 (Hölder spaces).

2.12 (Morrey inequality). (a) If $\frac{k}{d} < \frac{s}{d} - \frac{1}{p}$, then

$$\|\nabla^\alpha u\|_{C_0(\mathbb{R}^d)} \lesssim \|u\|_{W^{s,p}(\mathbb{R}^d)}, \quad u \in W^{s,p}(\mathbb{R}^d).$$

2.13 (Poincaré inequality). BMO

2.14 (Rellich-Kondrachov theorem). Let Ω be bounded open subset of $\mathbb{R}^d$ with Lipschitz boundary. For $1 \leq p < d$, p^* is given by $-\frac{1}{p^*} := \frac{1}{d} - \frac{1}{p}$, called the *Sobolev conjugate*. Let η_ε be a standard mollifier.

(a) The convolution operator $(\eta_\varepsilon * -) : L^1(\Omega) \rightarrow C(\overline{\Omega})$ is compact for each $\varepsilon > 0$.

(b) We have

$$\|\eta_\varepsilon * u - u\|_{L^1(\Omega)} \lesssim \varepsilon \|u\|_{W^{1,1}(\Omega)}, \quad u \in W^{1,1}(\Omega).$$

(c) If $1 \leq p < d$ and $1 \leq q < p^*$, then there is $\theta > 0$ such that we have

$$\|\eta_\varepsilon * u - u\|_{L^q(\Omega)} \lesssim \varepsilon^\theta \|u\|_{W^{1,p}(\Omega)}, \quad u \in W^{1,p}(\Omega).$$

(d) If $1 \leq p < d$ and $1 \leq q < p^*$, then the embedding $W^{1,p}(\Omega) \hookrightarrow L^q(\Omega)$ is compact.

(e) If $\frac{l}{d} - \frac{1}{q} < \frac{k}{d} - \frac{1}{p}$, then the embedding $W^{k,p}(\Omega) \hookrightarrow W^{l,q}(\Omega)$ is a compact.

Sketch. Take an approximate unit $e_n \in C_c^1(\mathbb{R}^d)$ of $L^1(\mathbb{R}^d)$ and let $T_n := e_n * -$. First we prove

$$T_n : L^1(\mathbb{R}^d) \rightarrow C_0(\mathbb{R}^d)$$

is compact for each n using the Arzela-Ascoli theorem. Then, the composition

$$W^{1,1}(\mathbb{R}^d) \rightarrow L^1(\mathbb{R}^d) \xrightarrow{T_n} C_0(\mathbb{R}^d) \rightarrow C_0(\Omega)$$

is also compact. Since Ω is bounded, we have $C_0(\Omega) \rightarrow L^1(\Omega)$. Now we have

$$T_n : W^{1,1}(\mathbb{R}^d) \rightarrow L^1(\Omega)$$

that is compact for each n . We can show that it converges to the embedding in norm by some estimates, so the restriction $W^{1,1}(\mathbb{R}^d) \rightarrow L^1(\Omega)$ is compact. Since Ω has the Lipschitz boundary, we have $W^{1,1}(\Omega) \rightarrow W^{1,1}(\mathbb{R}^d)$. Therefore, the embedding $W^{1,1}(\Omega) \rightarrow L^1(\Omega)$ is compact. $\square$

Proof. (a) The sequence $(\eta_\varepsilon * u_n)_n$ is pointwise bounded from

$$\|\eta_\varepsilon * u_n\|_{C_0(\mathbb{R}^d)} \leq \|\eta_\varepsilon\|_{C_0(\mathbb{R}^d)} \|u_n\|_{L^1(\mathbb{R}^d)} \lesssim 1, \quad n \in \mathbb{N},$$

and equicontinuous from

$$\|\nabla \eta_\varepsilon * u_n\|_{C_0(\mathbb{R}^d)} \leq \|\nabla \eta_\varepsilon\|_{C_0(\mathbb{R}^d)} \|u_n\|_{L^1(\mathbb{R}^d)} \lesssim 1, \quad n \in \mathbb{N}.$$

By the Arzela-Ascoli theorem, since $\overline{\Omega}$ is compact, there is a subsequence $(\eta_\varepsilon * u_{n_k})_k$ that is Cauchy in $C(\overline{\Omega})$.

(b) Write

$$\begin{aligned} \eta_\varepsilon * u(x) - u(x) &= \int \varepsilon^{-d} \eta(\varepsilon^{-1}(x-y))(u(y) - u(x)) dy \\ &= \int \eta(y)(u(x - \varepsilon y) - u(x)) dy \\ &= \int \eta(y) \int_0^1 \frac{d}{dt} (u(x - t\varepsilon y)) dt dy \\ &= - \int \varepsilon y \eta(y) \int_0^1 \nabla u(x - t\varepsilon y) dt dy. \end{aligned}$$

Then, since $|y| \leq 1$ if $\eta(y) > 0$,

$$\|\eta_\varepsilon * u - u\|_{L^1(\Omega)} \leq \varepsilon \int \eta(y) \int_0^1 \int |\nabla u(x - t\varepsilon y)| dx dt dy = \varepsilon \|\nabla u\|_{L^1(\mathbb{R}^d)}.$$

(c) Consider the interpolation

$$\|\eta_\varepsilon * u_n - u_n\|_{L^q(\Omega)} \leq \|\eta_\varepsilon * u_n - u_n\|_{L^1(\Omega)}^\theta \|\eta_\varepsilon * u_n - u_n\|_{L^{p^*}(\Omega)}^{1-\theta}$$

for $\frac{1}{q} = \frac{\theta}{1} + \frac{1-\theta}{p^*}$ with $0 < \theta \leq 1$. Since the Gagliardo-Nirenberg-Sobolev inequality gives the bound

$$\|\eta_\varepsilon * u_n - u_n\|_{L^{p^*}(\Omega)} \lesssim \|\eta_\varepsilon * u_n - u_n\|_{W^{1,p}(\Omega)} \lesssim 1, \quad n \in \mathbb{N}, \varepsilon > 0,$$

$$\sup_n \|\eta_\varepsilon * u_n - u_n\|_{L^q(\Omega)} \rightarrow 0$$

as $\varepsilon \rightarrow 0$.

(d) By the part (c), for any $\delta > 0$, there is $\varepsilon > 0$ such that

$$\sup_n \|\eta_\varepsilon * u_n - u_n\|_{L^q(\Omega)} < \frac{\delta}{2},$$

so for a subsequence $(\eta_\varepsilon * u_{n_k})_k$ that is Cauchy in $L^q(\Omega)$, we have

$$\|u_{n_k} - u_{n_{k'}}\|_{L^q(\Omega)} \leq \|\eta_\varepsilon * u_{n_k} - \eta_\varepsilon * u_{n_{k'}}\|_{L^q(\Omega)} + \delta,$$

and by the diagonal argument reducing δ to zero, we can construct the desired subsequence.

(e)

□

Chapter 3

Generalizations of Sobolev spaces

3.1 Fractional Sobolev spaces

3.2 Fourier transform methods

3.3 Almost everywhere differentiability

Lipschitz, Rademacher

Part II

Elliptic equations

Chapter 4

Potential theory

4.1 Mean value property

mean value property maximum principle Harnack inequality
potential estimate Hölder estimate

4.2 Weyl's lemma

Exercises

Problems

1. Let $d \geq 3$. Let u be a distribution on $\mathbb{R}^d$ that is harmonic on $\mathbb{R}^d \setminus \{0\}$ and vanishes at infinity. Then,
 $u = a_\alpha \partial^\alpha \Phi$.

Chapter 5

Existence theory

5.1 Variational methods

5.2 Lax-Milgram theorem

5.1. Let $L : H \rightarrow H$ be a densely defined linear operator. If there is a Hilbert space V containing $\text{dom } L$ and densely embedded in H such that $(u, v) \mapsto \langle Lu, v \rangle_H$ defines a coercive bilinear form on V , then L admits a surjective closure.

Proof. For $f \in H$, there is $v \in V$ such that $\langle f, \varphi \rangle_H = \langle v, \varphi \rangle_V$ for all $\varphi \in V$. If we let $u := A^{-1}v$, where $A \in B(V)$ is defined such that $\langle L-, - \rangle_H = \langle A-, - \rangle_V$. Then,

$$\langle Lu, \varphi \rangle_H = \langle Au, \varphi \rangle_V = \langle v, \varphi \rangle_V = \langle f, \varphi \rangle_H$$

implies $Lu = f$. □

5.2 (Poisson equation). Let Ω be a bounded open subset of $\mathbb{R}^d$. Consider the problem

$$\begin{cases} -\Delta u(x) = f(x) & , \text{ in } x \in \Omega, \\ u(x) = 0 & , \text{ on } x \in \partial\Omega. \end{cases}$$

Define a bilinear form B on $H_0^1(\Omega)$ such that

$$B(u, v) := \int \nabla u(x) \cdot \nabla v(x) dx.$$

- (a) If $u \in H_0^1(\Omega)$ and $f \in \mathcal{D}'(\Omega)$ satisfy $B(u, \varphi) = \langle f, \varphi \rangle$ for all $\varphi \in \mathcal{D}(\Omega)$, then $-\Delta u = f$.
- (b) B is another inner product equivalent to $\langle -, - \rangle_{H_0^1(\Omega)}$.
- (c) For $f \in H^{-1}(\Omega)$, there is $u \in H_0^{-1}(\Omega)$ such that $-\Delta u = f$.

5.3 Fredholm alternative

5.4 Perron's method

5.5 Eigenvalue problems

Chapter 6

Elliptic regularity

6.1 L^p theory

6.1 (Interior regularity in H^2). Let Ω be bounded open subset of $\mathbb{R}^d$ and $L : \mathcal{D}'(\Omega) \rightarrow \mathcal{D}'(\Omega)$ a uniformly elliptic operator given by

$$Lu := -\partial_j(a^{ij}\partial_i u) + b^i\partial_i u + cu$$

for $a^{ij} \in C^1(\Omega)$, $b^i \in L^\infty(\Omega)$, and $c \in L^\infty(\Omega)$.

Fix an open subset $U \Subset \Omega$ and $\zeta \in C_c^\infty(\Omega)$ a cutoff function such that $\zeta = 1$ in U . Let $\varphi := -\partial_k^{-h}(\zeta^2 \partial_k^h u)$ for $k = 1, \dots, d$ and sufficiently small $h > 0$.

(a) We have

$$\|\nabla u\|_{L^2(U)} \lesssim \|Lu\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)}$$

for all u such that $Lu, u \in L^2(\Omega)$

(b) We have

$$\int \zeta^2 |\partial_k^h \nabla u|^2 \lesssim \int a^{ij} \partial_i u \partial_j \varphi + \|\nabla u\|_{L^2(\Omega)}$$

for all $u \in H^1(\Omega)$.

(c) We have

$$\int \zeta^2 |\partial_k^h \nabla u|^2 \lesssim \|Lu\|_{L^2(\Omega)} + \|u\|_{H^1(\Omega)}$$

for all u such that $Lu \in L^2(\Omega)$ and $u \in H^1(\Omega)$.

(d) We have

$$\|u\|_{H^2(U)} \lesssim \|Lu\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)}$$

for all u such that $Lu, u \in L^2(\Omega)$.

Proof. (a) Since $\zeta^2 u \in H_0^1(\Omega)$,

$$\begin{aligned}
\int \zeta^2 |\nabla u|^2 &\lesssim \int a^{ij} \zeta^2 \partial_i u \partial_j u \\
&= \int a^{ij} \partial_i u \partial_j (\zeta^2 u) - \int a^{ij} \partial_i u \partial_j (\zeta^2) u \\
&= \int (Lu - b^i \partial_i u - cu) \zeta^2 u - \int a^{ij} \partial_i u 2\zeta \partial_j \zeta u \\
&\lesssim \int (|Lu| + |u| \zeta |\nabla u| + |u|^2 + |u| \zeta |\nabla u|) \\
&\lesssim \int (|Lu|^2 + |u|^2) + \frac{1}{\varepsilon} \int |u|^2 + \varepsilon \int \zeta^2 |\nabla u|^2.
\end{aligned}$$

Taking small $\varepsilon > 0$, we are done.

(b) Write

$$\begin{aligned}
\int a^{ij} \partial_i u \partial_j \varphi &= - \int a^{ij} \partial_i u \partial_j \partial_k^{-h} (\zeta^2 \partial_k^h u) \\
&= \int \partial_k^h (a^{ij} \partial_i u) \partial_j (\zeta^2 \partial_k^h u) \\
&= \int \partial_k^h a^{ij} \partial_i u \partial_j (\zeta^2) \partial_k^h u + \int \partial_k^h a^{ij} \partial_i u \zeta^2 \partial_j \partial_k^h u \\
&\quad + \int a^{ij} \partial_k^h \partial_i u \partial_j (\zeta^2) \partial_k^h u + \int a^{ij} \partial_k^h \partial_i u \zeta^2 \partial_j \partial_k^h u.
\end{aligned}$$

The last term out of the four terms controls the difference quotient $|\partial_k^h \nabla u|$ as

$$\int a^{ij} \partial_k^h \partial_i u \zeta^2 \partial_j \partial_k^h u \gtrsim \int \zeta^2 |\partial_k^h \nabla u|^2,$$

and the absolute values of other three terms are estimated up to constant by

$$\begin{aligned}
&\int \zeta |\nabla u| |\partial_k^h u| + \int \zeta^2 |\nabla u| |\partial_k^h \nabla u| + \int \zeta |\partial_k^h \nabla u| |\partial_k^h u| \\
&\lesssim \left(1 + \frac{1}{\varepsilon}\right) \int \zeta^2 |\nabla u|^2 + \left(1 + \frac{1}{\varepsilon}\right) \int |\partial_k^h u|^2 + \varepsilon \int \zeta^2 |\partial_k^h \nabla u|^2 \\
&\lesssim \left(1 + \frac{1}{\varepsilon}\right) \int |\nabla u|^2 + \varepsilon \int \zeta^2 |\partial_k^h \nabla u|^2.
\end{aligned}$$

Therefore,

$$\int \zeta^2 |\partial_k^h \nabla u|^2 \lesssim \int a^{ij} \partial_i u \partial_j \varphi + \left(1 + \frac{1}{\varepsilon}\right) \int |\nabla u|^2 + \varepsilon \int \zeta^2 |\partial_k^h \nabla u|^2,$$

and taking small $\varepsilon > 0$, we are done.

(c) Note that

$$\int a^{ij} \partial_i u \partial_j \varphi = \int (Lu - b^i \partial_i u - cu) \varphi$$

since $\varphi \in H_0^1(\Omega)$. Because

$$\int (Lu - b^i \partial_i u - cu) \varphi \lesssim \frac{1}{\varepsilon} \int (|Lu|^2 + |\nabla u|^2 + |u|^2) + \varepsilon \int |\varphi|^2$$

and

$$\begin{aligned}
\int |\varphi|^2 &= \int |\partial_k^{-h}(\zeta^2 \partial_k^h u)|^2 \\
&\lesssim \int |\nabla(\zeta^2 \partial_k^h u)|^2 \\
&\lesssim \int |\partial_k^h u|^2 + \int \zeta^2 |\partial_k^h \nabla u|^2 \\
&\lesssim \int |\nabla u|^2 + \int \zeta^2 |\partial_k^h \nabla u|^2,
\end{aligned}$$

we obtain

$$\int (Lu - b^i \partial_i u - cu) \varphi \lesssim \frac{1}{\varepsilon} \int (|Lu|^2 + |u|^2) + \left(\varepsilon + \frac{1}{\varepsilon} \right) \int |\nabla u|^2 + \varepsilon \int \zeta^2 |\partial_k^h \nabla u|^2.$$

Taking small $\varepsilon > 0$, we are done. □

6.2 Schauder theory

6.3 De Giorgi-Nash-Moser theory

6.4 Viscosity solutions

Part III

Evolution equations

Chapter 7

Parabolic equations

7.1 Galerkin approximation

7.2 Semigroup theory

Hille-Yosida Lumer-Phillips

C_0 -semigroup, analytic semigroup

Chapter 8

Hyperbolic equations

Chapter 9

Local and global existence

9.1 Local existence

contraction mapping

9.2 Global existence

a priori estimates gronwall inequality

9.3 Weak convergence

Part IV

Nonlinear equations

Chapter 10

Chapter 11

Hamilton-Jacobi equations

optimal control viscosity solution

Chapter 12

Conservation laws

shocks NS